

Deepak Mallampati

APPLIED AI ENGINEER - AGENTIC WORKFLOWS & AI TOOLING

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years building production agentic workflows, AI tooling, and automation on LLM platforms. Architect of a conductor-subagent orchestration framework (42% token reduction) and healthcare RAG pipelines reaching 92% precision with ~40% hallucination reduction. Hands-on with Python, tool-use/function-calling, vector databases (FAISS, Pinecone, Weaviate), orchestration layers, and content-safety guardrails (prompt-injection mitigation, PII redaction, RBAC) - exactly the stack for building and iterating agentic harnesses and internal automation in a security-conscious org.

CORE COMPETENCIES

Agentic Workflows • AI Tooling & Automation • RAG • Tool-Use / Function-Calling • Vector Databases • Orchestration Layers • Multi-Agent Systems • Python • Content Safety & Guardrails

Generative AI & LLMs: LangChain/LangGraph, RAG architecture, multi-agent systems, prompt engineering, tool-use/function-calling, GPT-4o/GPT-5, Claude, Gemini, LLaMA

Vector Search & Retrieval: FAISS, Pinecone, Weaviate, semantic & hybrid search, cross-encoder re-ranking, document chunking, semantic caching

Automation & Orchestration: scheduled batch orchestration, exponential-backoff retry, model routing, workflow automation, CI/CD (Jenkins, Azure DevOps)

Security & Compliance: prompt-injection mitigation, PII masking/redaction, RBAC, OAuth 2.0/JWT, audit logging, zero-trust, HIPAA/SOC 2/PCI DSS

Languages & Tools: Python, JavaScript/TypeScript, SQL, FastAPI, Git; PostgreSQL, MongoDB, Redis; Docker, Kubernetes

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing objectives into context-scoped subagents executing autonomously - **42%** token reduction at sustained accuracy across multi-step workflows.
- Authored and iterated **agentic harnesses and AI tooling** with tool-use/function-calling and model routing, plus fault-tolerant automation infrastructure (scheduled batch orchestration, exponential-backoff retry) that cut failed production runs by **60%**.
- Engineered **RAG pipelines** over large healthcare corpora (dense retrieval + cross-encoder re-ranking), lifting answer relevance and materially reducing hallucination on domain-specific queries.
- Built privacy-preserving workflows (K-anonymity, L-diversity, differential privacy) and an **LLM evaluation framework** (RAGAS, BERTScore) with latency/accuracy dashboards.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated GPT-4o, Claude, and Gemini, improving task success by **27%**; engineered content-safety controls reducing unsafe outputs by **42%**.
- Built **12 APIs and microservices**, cutting p95 latency 1.5s → **800ms** and supporting **100,000+** requests daily across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline on clinical data; reduced re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned Qwen3 via 4-bit QLoRA on an H100; authored an IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Built regression/classification models for failure prediction, anomaly detection, and forecasting; fine-tuned BERT/RoBERTa for NER reaching **89% F1**.

PROJECTS

career-ops

An autonomous, agentic AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - internal automation and ongoing experimentation with emerging AI in one shipped system.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - single-action inline tooling built end to end.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0